

# Module 4: Access Files and Folders

## Exercise 1: Applying Permissions and Accessing Files

In this exercise, you apply various permissions and review the effect on file access.

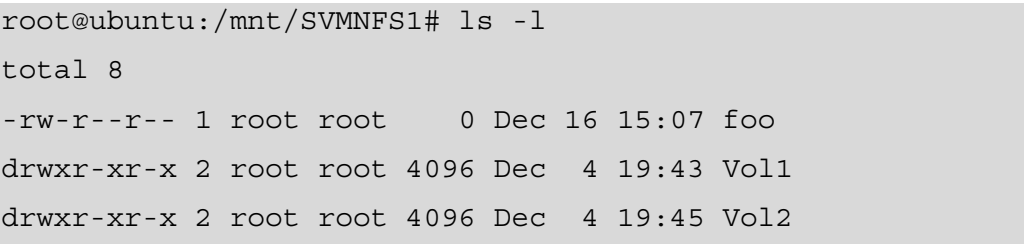
### Objectives

This exercise focuses on enabling you to do the following:

- Create files in a storage VM (storage virtual machine, also known as SVM) namespace directory
- Modify mode bits and review its effect on file access
- Change the export policy to read-only and examine its effect on file access

### Task 1: Access the Files and Folders

In this task, you access the various directories and create files.

| Step | Action                                                                                                                                                                                                                                                                         |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1-1  | In the Ubuntu client CLI, change the directory to the mount location.<br><code>cd SVMNFS1</code>                                                                                                                                                                               |
| 1-2  | List the directory contents.<br><code>ls</code>                                                                                                                                                                                                                                |
| 1-3  | Create a file in the root directory of the storage VM namespace.<br><code>touch foo</code>                                                                                                                                                                                     |
| 1-4  | List the directory contents.<br><code>ls -l</code><br>                                                                                                                                     |
| 1-5  |  The NetApp recommended practice is <i>not</i> to create files in the storage VM namespace root. In this exercise, files are created in the namespace root for demonstration purposes only. |
| 1-6  | Change the directory to <b>Vol1</b> .<br><code>cd Vol1</code>                                                                                                                                                                                                                  |
| 1-7  | Create a file.<br><code>touch foo</code>                                                                                                                                                                                                                                       |

| Step | Action                                                        |
|------|---------------------------------------------------------------|
| 1-8  | List the directory contents.<br><code>ls</code>               |
| 1-9  | Change the directory to one level up.<br><code>cd ..</code>   |
| 1-10 | Change the directory to <b>Vol2</b> .<br><code>cd Vol2</code> |
| 1-11 | Create a file.<br><code>touch foo</code>                      |
| 1-12 | List the directory contents.<br><code>ls</code>               |

## Task 2: Modify Mode Bits and Review the Effects on File Permissions

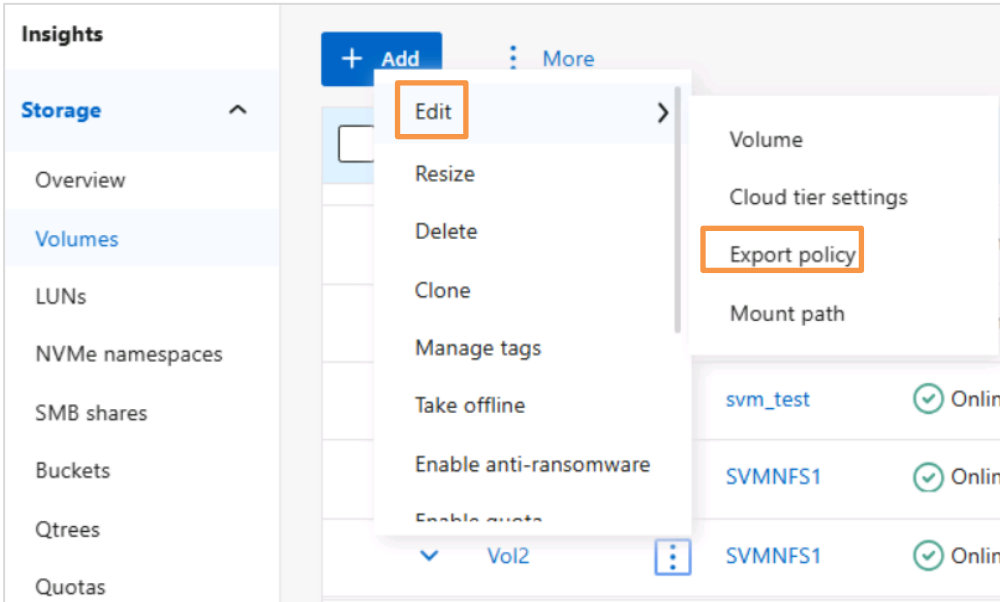
In this task, you modify the mode bits for the student user and review the effect on the file access permissions.

| Step | Action                                                                                                                                                                                                                                                                                                                       |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2-1  | Examine the access permissions for the file <b>foo</b> .<br><code>ls -l foo</code><br><pre>root@ubuntu:/mnt/SVMNFS1/Vol2# ls -l foo -rw-r--r-- 1 root root 0 Dec 16 15:14 foo</pre>                                                                                                                                          |
| 2-2  |  The file <b>foo</b> is an empty file (0 bytes).<br>It is owned by user <b>root</b> and group <b>root</b> .                                                                                                                               |
| 2-3  | Observe the following set of permissions (UNIX mode bits): <ul style="list-style-type: none"> <li><b>rw-</b> (user)—<b>root</b> is the owner, and it can read and write but cannot execute</li> <li><b>r--</b> (group)—group name is <b>root</b>, and it can only read</li> <li><b>r--</b> (others)—can only read</li> </ul> |
| 2-4  | Open a new Ubuntu client session and log in as the <b>student</b> user.                                                                                                                                                                                                                                                      |
| 2-5  | Enter <b>NetApp1!</b> as the password.                                                                                                                                                                                                                                                                                       |
| 2-6  | Change the directory to the <b>Vol2</b> folder.<br><code>cd /mnt/SVMNFS1/Vol2</code>                                                                                                                                                                                                                                         |
| 2-7  | Edit the file <b>foo</b> .<br><code>echo "This is my NFS file" &gt;foo</code>                                                                                                                                                                                                                                                |

| Step | Action                                                                                                                                     |
|------|--------------------------------------------------------------------------------------------------------------------------------------------|
| 2-8  | Answer the following questions: <ul style="list-style-type: none"> <li>Were you able to edit the file?</li> <li>Why or why not?</li> </ul> |
| 2-9  | Return to the Ubuntu session that is logged in as the root user.                                                                           |
| 2-10 | Change the mode bit permission for others to read and write.<br><code>chmod o+rw foo</code>                                                |
| 2-11 | Edit the file from the Ubuntu session of the student user:<br><code>echo "This is my NFS file" &gt;foo</code>                              |
| 2-12 | Were you able to edit the file this time?                                                                                                  |

### Task 3: Change the Export Policy for a Volume and Review the Effect

In this task, you change the export policy for Vol2 and review the effect on file access permissions.

| Step | Action                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3-1  | From the NetApp ONTAP System Manager menu, select <b>Storage &gt; Volumes</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 3-2  | Under Volumes, click the <b>ellipsis (three vertical dots)</b> that correspond to Vol2.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 3-3  | Select <b>Edit &gt; Export policy</b> .<br> <p>The screenshot shows the NetApp ONTAP System Manager interface. On the left, the 'Storage' menu is expanded, showing 'Volumes' selected. In the main area, a list of volumes is displayed, including 'Vol2'. A context menu is open for 'Vol2', showing options like 'Edit', 'Resize', 'Delete', 'Clone', 'Manage tags', 'Take offline', 'Enable anti-ransomware', and 'Enable quota'. The 'Edit' option is highlighted, and a sub-menu is visible with 'Export policy' selected. Below the menu, a table shows the status of various export policies: 'svm_test' (Online), 'SVMNFS1' (Online), and 'SVMNFS1' (Online).</p> |

| Step       | Action                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                   |                |                  |                  |                 |                |   |           |                   |     |       |                  |   |           |                   |     |     |     |
|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|----------------|------------------|------------------|-----------------|----------------|---|-----------|-------------------|-----|-------|------------------|---|-----------|-------------------|-----|-----|-----|
| 3-4        | <p>In the Edit export policy dialog box, change the Export policy value to <b>ExportPolicy1</b>.</p> <div><div><div>Edit export policy</div><div><div>Volume</div><div>Vol2</div></div><div><div>Mount path</div><div>/Vol2</div></div><div><div>Export policy</div><div><div><div><div></div></div>Select an existing policy</div><div><div>Export policy</div><div>ExportPolicy1</div></div></div></div><div><div>Rules</div><table><tr><th>Rule index</th><th>Clients</th><th>Access protocols</th><th>Read-only rule</th><th>Read/write rule</th><th>SuperUser acco</th></tr><tr><td>1</td><td>0.0.0.0/0</td><td>NFSv3, NFSv4, NFS</td><td>Any</td><td>Never</td><td>Krb5, Krb5i, Krb</td></tr><tr><td>2</td><td>0.0.0.0/0</td><td>NFSv3, NFSv4, NFS</td><td>Any</td><td>Any</td><td>Any</td></tr></table></div></div></div> | Rule index        | Clients        | Access protocols | Read-only rule   | Read/write rule | SuperUser acco | 1 | 0.0.0.0/0 | NFSv3, NFSv4, NFS | Any | Never | Krb5, Krb5i, Krb | 2 | 0.0.0.0/0 | NFSv3, NFSv4, NFS | Any | Any | Any |
| Rule index | Clients                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Access protocols  | Read-only rule | Read/write rule  | SuperUser acco   |                 |                |   |           |                   |     |       |                  |   |           |                   |     |     |     |
| 1          | 0.0.0.0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | NFSv3, NFSv4, NFS | Any            | Never            | Krb5, Krb5i, Krb |                 |                |   |           |                   |     |       |                  |   |           |                   |     |     |     |
| 2          | 0.0.0.0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | NFSv3, NFSv4, NFS | Any            | Any              | Any              |                 |                |   |           |                   |     |       |                  |   |           |                   |     |     |     |
| 3-5        | Click <b>Save</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                   |                |                  |                  |                 |                |   |           |                   |     |       |                  |   |           |                   |     |     |     |
| 3-6        | <p>In the Ubuntu root user session, create a file.</p> <pre>touch filenew</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                   |                |                  |                  |                 |                |   |           |                   |     |       |                  |   |           |                   |     |     |     |
| 3-7        | <p>Answer the following questions:</p> <ul style="list-style-type: none"><li>• Were you able to create a file?</li><li>• Why or why not?</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                   |                |                  |                  |                 |                |   |           |                   |     |       |                  |   |           |                   |     |     |     |
| 3-8        | Return to ONTAP System Manager and change the export policy to <b>default</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                   |                |                  |                  |                 |                |   |           |                   |     |       |                  |   |           |                   |     |     |     |

**End of exercise**